

CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

**A Project Report Submitted in Partial Fulfillment of the Requirements for
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ABSTRACT:

Customer churn prediction is a critical problem in many industries, where retaining customers is more cost-effective than acquiring new ones. This project aims to develop a machine learning model that predicts whether a customer is likely to churn based on historical data such as demographics, service usage, and billing information. Various data preprocessing techniques, exploratory data analysis, and machine learning algorithms such as Logistic Regression, Decision Trees, and Random Forest were applied. The model was evaluated using performance metrics like accuracy, precision, recall, and F1-score. The results show that the model can effectively identify potential churn customers, enabling businesses to take proactive measures for customer retention.

1. INTRODUCTION

In today's highly competitive digital economy, businesses rely heavily on customer retention for sustained growth and profitability. Customer churn, also known as customer attrition, occurs when customers stop using a company's products or services. This is a major concern because acquiring new customers is significantly more expensive than retaining existing ones.

Industries such as telecommunications, banking, SaaS platforms, and e-commerce face high churn rates due to intense competition and evolving customer expectations. Therefore, organizations aim to predict churn before it happens so they can take proactive actions like offering discounts, improving services, or personalized engagement.

This project focuses on applying machine learning techniques to predict customer churn using historical customer data. The system analyzes patterns in customer behavior, usage trends, and billing information to determine the likelihood of churn.

The significance of this project lies in its ability to bridge data analysis and business decision-making. By leveraging predictive analytics, companies can reduce churn rates, improve customer satisfaction, and maximize revenue.

2. OBJECTIVES

The project is designed with both technical and business objectives:

Primary Objectives:

- To develop a predictive model that accurately classifies customers as churn or non-churn
- To identify key factors that influence customer churn
- To improve decision-making through data-driven insights

Secondary Objectives:

- To gain hands-on experience with real-world datasets
- To understand the complete machine learning pipeline
- To implement data preprocessing and feature engineering techniques

- To compare multiple machine learning models and select the best one
- To build a deployable system for real-time predictions

This project also aims to strengthen core skills such as data analysis, model evaluation, and problem-solving.

3. PROBLEM STATEMENT

Customer churn is a critical business issue that directly impacts revenue and long-term sustainability. Companies often lose customers due to poor service, high costs, or better competitor offerings.

The challenge is not just identifying churn after it happens, but predicting it in advance. This is difficult because:

- Customer behavior is complex and influenced by multiple factors
- The dataset may contain noisy or missing data
- Churn cases are often fewer than non-churn cases (class imbalance problem)
- Relationships between variables may be non-linear

The goal of this project is to build a classification model that predicts whether a customer will churn based on input features. The output is binary:

- **1 (Churn)** – Customer is likely to leave
- **0 (No Churn)** – Customer is likely to stay

A successful solution should be accurate, reliable, and interpretable.

4. DATASET DESCRIPTION

The dataset used in this project represents customer information collected over time. It contains both categorical and numerical features.

Types of Data:

1. Demographic Information:

- Gender
- Senior Citizen status
- Dependents

2. Account Information:

- Tenure (number of months with company)
- Contract type (monthly, yearly)

3. Service Details:

- Internet service type
- Phone service
- Additional services like streaming, security, etc.

4. Billing Information:

- Monthly charges
- Total charges
- Payment method

5. Target Variable:

- Churn (Yes/No)

Dataset Characteristics:

- Mixed data types (categorical + numerical)
- Potential missing values
- Imbalanced classes

Understanding the dataset is crucial before applying any machine learning techniques.

5. DATA PREPROCESSING

Data preprocessing is one of the most important steps in the machine learning pipeline. Poor-quality data leads to poor model performance.

5.1 Handling Missing Values

Missing values can distort model predictions. In this project:

- Rows with missing critical values were removed
- Alternatively, missing values were imputed using mean/median

5.2 Data Cleaning

- Removed duplicate records
- Fixed inconsistent data formats
- Converted numerical columns stored as strings into proper numeric types

5.3 Encoding Categorical Variables

Machine learning models require numerical input. Therefore:

- Binary categories were label encoded (0/1)
- Multi-class variables were one-hot encoded

5.4 Feature Scaling

Scaling ensures all features contribute equally:

- Standardization (mean = 0, std = 1)
- Prevents bias toward large-value features

5.5 Handling Imbalanced Data

- Techniques like oversampling or undersampling can be used
- Helps improve model performance on minority class (churn)

6. EXPLORATORY DATA ANALYSIS (EDA)

EDA is used to extract insights and understand patterns in the dataset.

6.1 Distribution Analysis

- Visualized churn vs non-churn ratio
- Identified imbalance in target variable

6.2 Feature-wise Analysis

Tenure vs Churn

- Customers with low tenure are more likely to churn
- Long-term customers show loyalty

Monthly Charges vs Churn

- Higher charges → higher probability of churn

Contract Type

- Monthly contracts have highest churn
- Long-term contracts reduce churn

Internet Service

- Fiber optic users showed higher churn rates

6.3 Correlation Analysis

- Used correlation heatmaps to identify important features
- Helped in feature selection

6.4 Visualization Tools Used

- Bar plots
- Histograms
- Heatmaps
- Box plots

7. MODEL SELECTION

Choosing the right algorithm is critical.

7.1 Logistic Regression

- Used for binary classification
- Provides probability output
- Easy to interpret

7.2 Decision Tree

- Splits data based on feature values
- Easy to visualize
- Can overfit if not controlled

7.3 Random Forest

- Ensemble of multiple decision trees
- Reduces overfitting
- Provides high accuracy

7.4 Support Vector Machine (SVM)

- Finds optimal boundary between classes
- Works well for complex datasets
- Requires more computation

Each model was tested and compared based on performance.

8. MODEL TRAINING

The dataset was divided into training and testing sets:

- **Training Set (80%)** – Used to train the model
- **Testing Set (20%)** – Used to evaluate performance

Steps in Training:

- Feature selection
- Model initialization
- Training on dataset
- Hyperparameter tuning using techniques like Grid Search

Overfitting Control:

- Used cross-validation
- Limited tree depth (for decision trees)

Training ensures the model learns patterns effectively.

9. MODEL EVALUATION

Evaluation is done using multiple metrics:

Accuracy

Measures overall correctness

Precision

Important when false positives are costly

Recall

Important for detecting actual churn customers

F1 Score

Balances precision and recall

Confusion Matrix

Provides detailed breakdown:

- True Positives
- False Positives

- True Negatives
- False Negatives

Observation:

Random Forest gave the best balance between accuracy and recall, making it suitable for churn prediction.

10. RESULTS AND DISCUSSION

Key Findings:

- Contract type strongly influences churn
- Customers with high bills tend to leave
- Short tenure is a major churn indicator

Model Performance:

- Random Forest achieved highest accuracy
- Logistic Regression provided interpretability
- Decision Tree was easy to visualize

Business Implications:

- Companies can identify high-risk customers
- Targeted marketing campaigns can be applied
- Improve service quality for at-risk segments

11. MODEL DEPLOYMENT

Deployment makes the model usable in real-world scenarios.

Tools Used:

- Flask / FastAPI for backend

- Streamlit for UI

Deployment Workflow:

1. User inputs customer details
2. Data is preprocessed
3. Model predicts churn probability
4. Result displayed on UI

Use Case:

Customer service teams can use this system to:

- Predict churn in real-time
- Offer retention strategies

12. TOOLS AND TECHNOLOGIES USED

Programming Language:

- Python

Libraries:

- Pandas (data manipulation)
- NumPy (numerical operations)
- Scikit-learn (ML models)
- Matplotlib & Seaborn (visualization)

Development Tools:

- VS Code

Version Control:

- Git and GitHub

13. CHALLENGES FACED

- Handling missing and inconsistent data
- Dealing with class imbalance
- Selecting the right features
- Avoiding overfitting
- Model tuning complexity

Each challenge was addressed using proper techniques like preprocessing, validation, and tuning.

14. FUTURE IMPROVEMENTS

- Use deep learning models (Neural Networks)
- Apply advanced feature engineering
- Use larger and real-time datasets
- Deploy on cloud (AWS, Azure)
- Integrate with business dashboards

15. CONCLUSION

This project demonstrates the practical application of machine learning in solving a real-world business problem. By analyzing customer data and building predictive models, churn can be identified in advance.

The model developed provides valuable insights into customer behavior and enables proactive decision-making. With further enhancements, this system can be scaled into a full production-level solution.